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AERO60492 – Autonomous Mobile Robots

Lecture 7 – Robotic Actuators and Sensors

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Today’s session

- 1 hour lecture
- 1 hour group activity

Intended learning outcomes

- Understand different types of actuators
- Explain the difference between servomotor and motor
- Identify different modes of locomotion and their basic characteristics
- Describe common sensor types used in robotics
- Argue pros and cons of using specific sensor

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Actuators overview

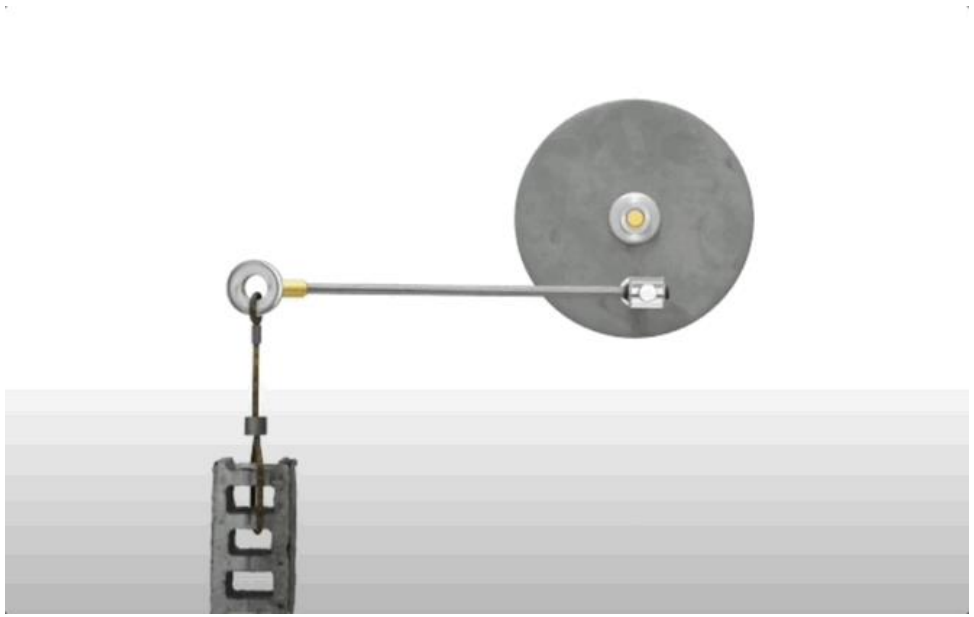
What is an actuator?

part of a machine or system that moves something or makes something work

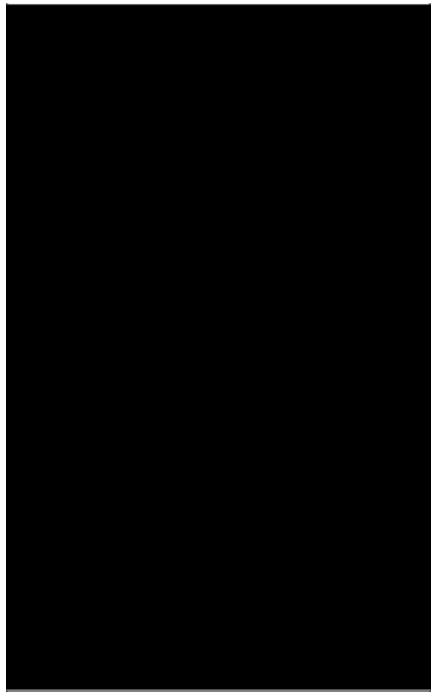
Cambridge dictionary

Basic actuator overview

- Rotary:



- Linear:



Actuators source of power

- Hydraulics/Pneumatics
- Electrics – most common
- Thermal – extreme rare
- Fuel/Gas – almost never used in robotics (indoor usage and maintenance)

What is a motor?

a device that changes electricity or fuel into movement

Cambridge dictionary

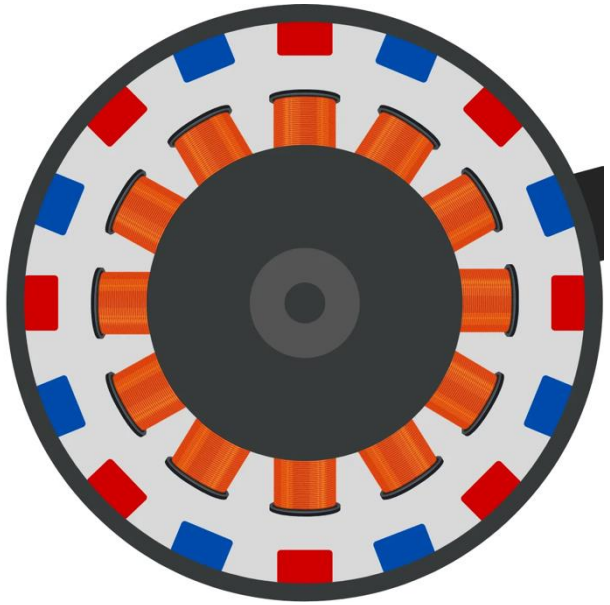
- Motor vs Engine:
  - By convention we use motor to mean electrical input and engine to mean petrol/hydrogen/combustible fluid input

Electrical

- By far most common in robotics
- Cheap and easy to maintain
- Commonly used for both rotary and linear actuation
- Fo rotary, generally two types of actuators: brushed and brushless motors

Brushed vs Brushless motors

Brushed	Brushless
Cheap	lighter
	More powerful
	More efficient
	Needs complex controller



Servomotors

- Servomotor allows precise control of rotation angle and velocity
- Unlike standard motors they can hold their position against external force for prolonged periods of time
- Several applications:
  - Ailerons on the aircraft
  - Legged robots
  - Robotic manipulators
- Programmable motors which can be either a servo or a motor

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Locomotion Methods

Locomotion methods

- Ground
  - Wheels
  - Omni wheels
  - Tracks
  - Legs
- Air
  - Propellers
  - Jet engines
- Water
  - Propellers
  - Fins
  - Water jets

We consider

- Ground
  - Wheels
  - Omni wheels
  - Tracks
  - Legs
- Air
  - Propellers
  - Jet engines
- Water
  - Propellers
  - Fins
  - Water jets

Wheeled

- Similar to cars, interact with ground using wheels
- Some important parameters:
  - Wheel diameter
  - Wheel width
  - Nominal tire pressure
  - Tire rolling resistance
- Notable features:
  - Very energy efficient
  - Struggles to climb larger obstacles, including stairs

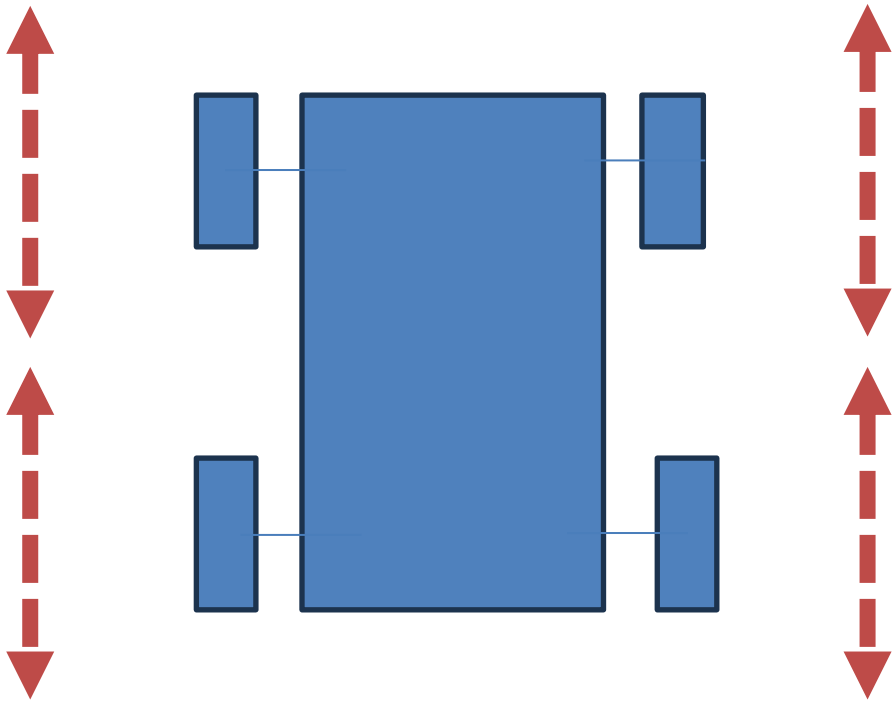
Omniwheeled

- Wheels which can move in any direction!
- Several kinds exist, here we briefly describe mecanum wheels:

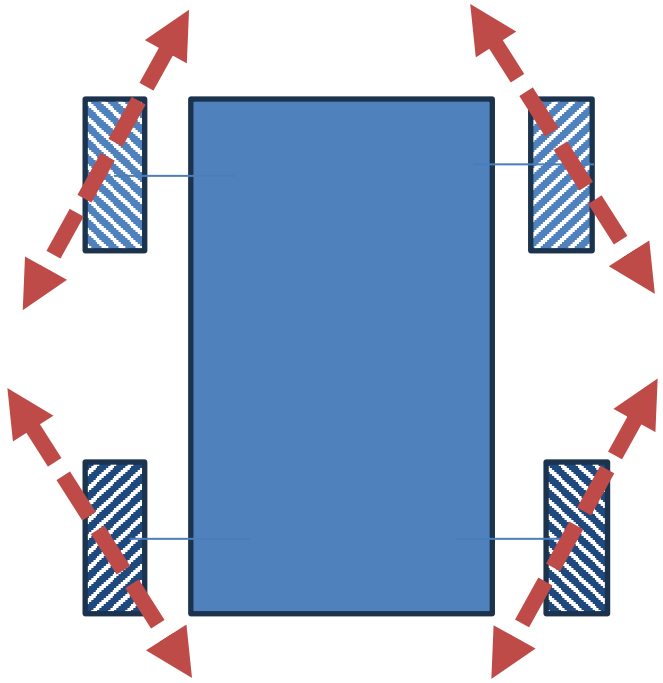
- They rely on very careful balance of friction between the wheels and fail to work on slippery surfaces

Omniwheeled – force direction

- Traditional wheel



- Mecanum wheel



Mecanum drive simulator

- https://seamonsters-2605.github.io/archive/mecanum/

Tracked

- Use tracks to move on terrain
- Notable features:
  - Great at traversing difficult terrain
  - Can climb taller obstacles
  - Does not climb stairs
  - Slow
  - Tracks are expensive and difficult to maintain
  - Once the track falls, the robot is stuck permanently

Legged

- Legged robots only recently became feasible
- Notable features:
  - Great at traversing difficult terrain
  - Can climb taller obstacles
  - Can be very fast
  - Can climb stairs
  - Extremely difficult and complex to control

Legged – why so useful?

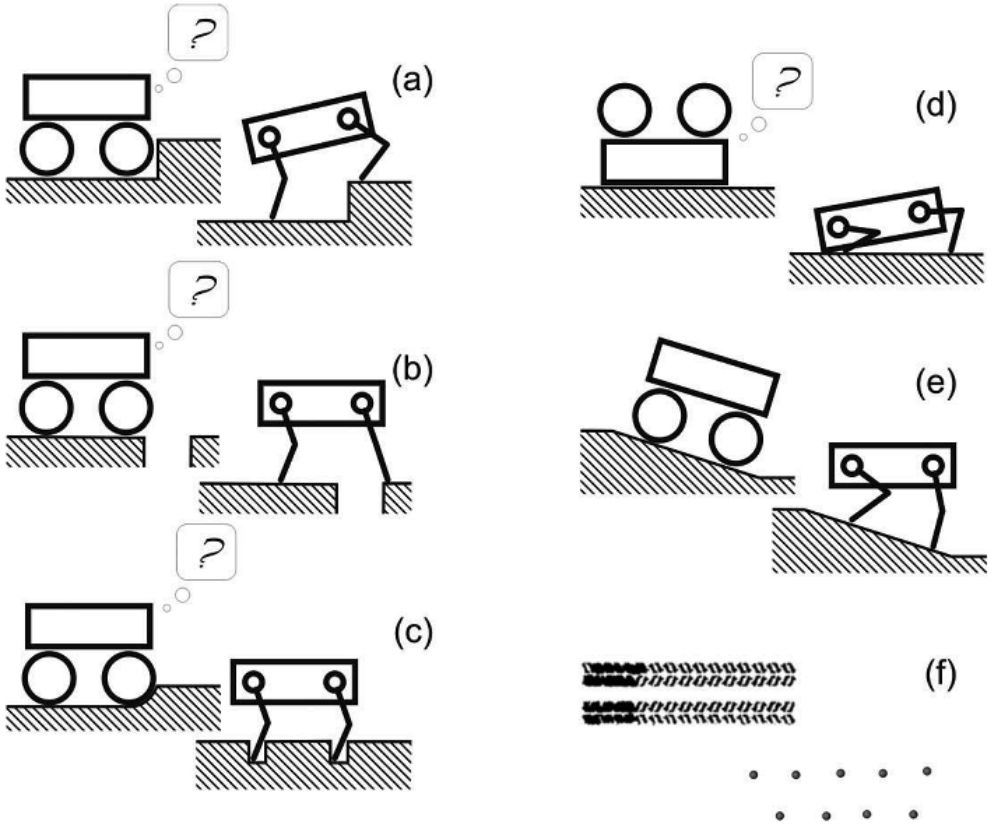


Figure 2.5 Legged robots are particularly suited for rough terrain, where they are able to traverse obstacles such as steps (a), gaps (b), or sandy patches (c) that are impassable for wheeled systems. Additionally, the high number of degrees of freedom allows the robot to stand up when fallen (d) and keep its payload leveled (e). Because legged systems do not require a continuous path for support, they can rely on a few selected footholds, which also reduces the environmental impact (f). Image courtesy of D. Remy.

Taken from “Introduction to Autonomous Mobile Robots”, R. Siegwart, I. R. Nourbakhsh and D. Scaramuzza.

Legged – why so difficult?

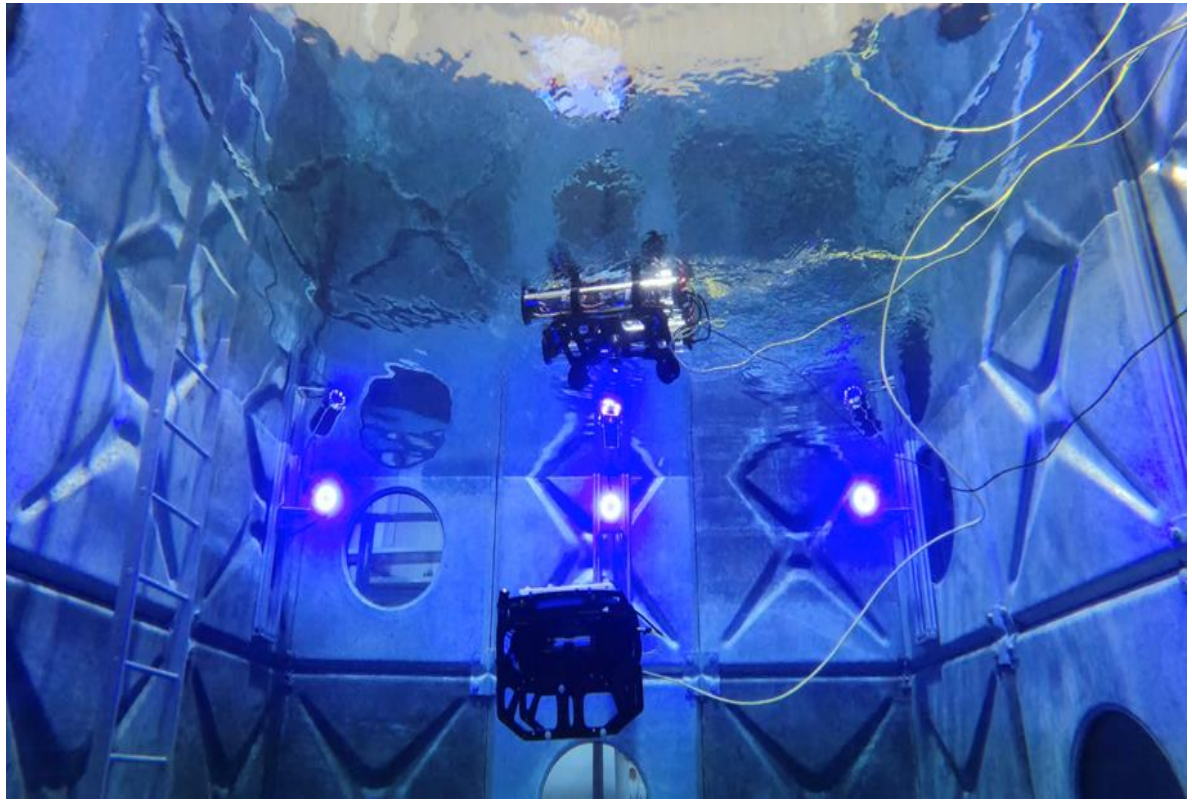
- It takes human a year to learn how to stand, another 6 months to walk
- Balancing (even in one pose) is extremely challenging (brain is doing amazing work and we are still not sure how)
- Actuators (until recently) were not accurate enough
- Boston Dynamics took almost 20 years to get technology ready for Spot – other companies are mostly using open source releases from Boston Dynamics

Aerial

- Propellers are used for majority of aerial robots
- They generate thrust which is used to keep the vehicles in the air
- Notable features
  - Do not care about terrain or stairs
  - Very limited mission time
  - Fragile so damage is to be avoided at all cost

Underwater

- Also use propellers, although working fluid is water, so designs are very different
- Notable features
  - Moving in water is energy expensive so they move slowly
  - Everything must be waterproof
  - It needs to be able to cope with water current so actuators must be powerful



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Sensors

Why do we need sensors?

- Robot needs to see:
  - Cameras
- Robot needs to navigate:
  - LIDAR
  - Depth Cameras
  - IMUs
  - Ultrasonic sensors
  - Encoders
- Robot needs to complete its objective
  - Pressure/temperature/light sensors
  - Thermal cameras

Sensors overview

Sensors overview

Depth camera/ camera

LIDAR

IMU

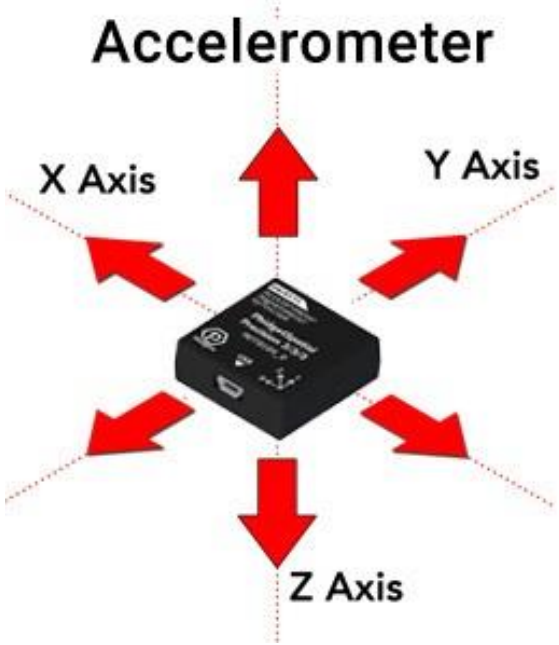
Mission sensors

Ultrasonic sensors

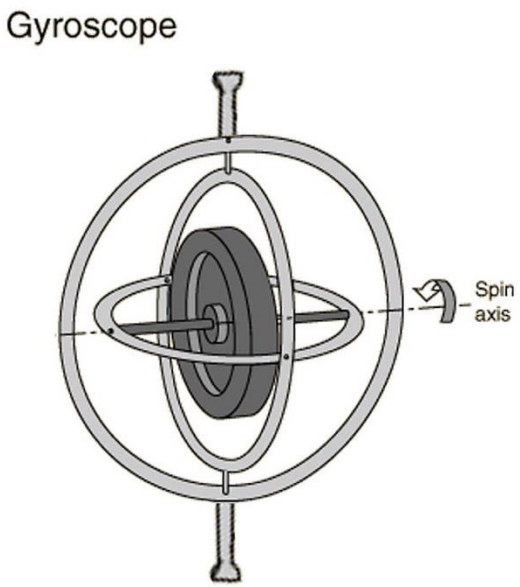
Encoders

Inertial Measurement Unit (IMUs)

- Consists of two/three sensors in each axis:



Measures acceleration in three directions



Measures rotations in three directions

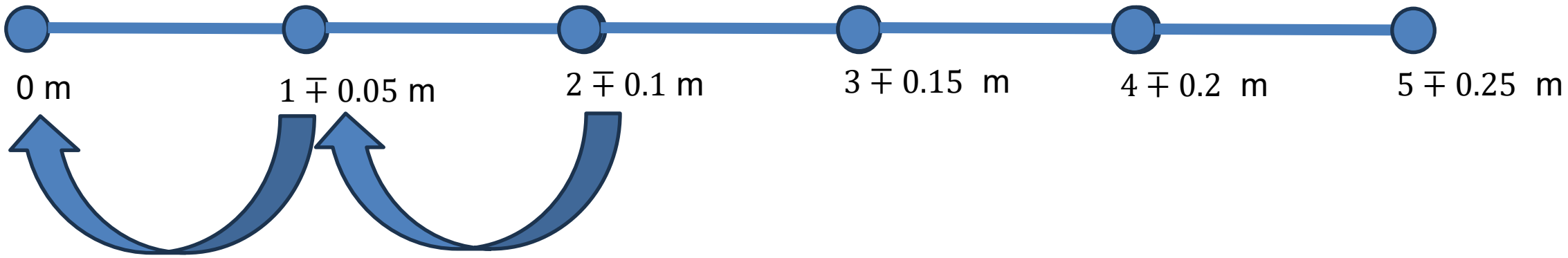
Measures magnetic field (magnetic north)

IMU pitfall- estimating position with just IMU

- It is done for commercial aviation – IMUs are very expensive and calibrated regularly

- Compounding errors! – when small errors accumulate to result in a larger error overall

- Imagine scenario:
  - an accelerometer which has 5% error for each 1 meter moved
  - Robot moves in 1 direction, and measures every 1 meter how far it moved



Compute how far it moved based on integration

Each integration introduces additional $\mp 0.05\ m$ error

Sensors overview

Depth camera/ camera

IMU

Encoders

LIDAR

Mission sensors

Ultrasonic sensors

Encoders

- Encoders allows us to determine how far motor has rotated
- Useful to provide estimation for speed and distance for ground vehicles
- Does not account for wheel to ground slippage thus not that accurate

Sensors overview

Depth camera/ camera

IMU

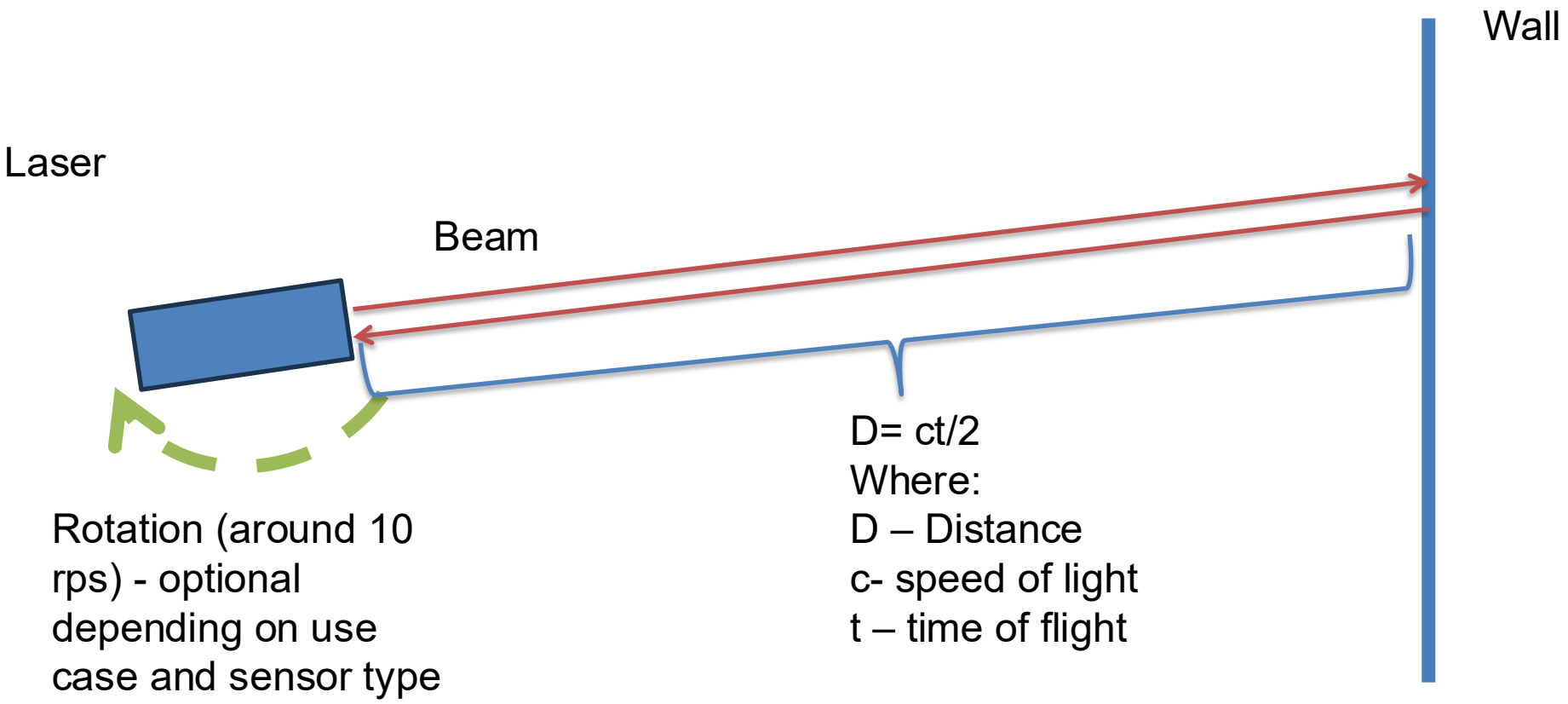
Encoders

LIDAR

Mission sensors

Ultrasonic sensors

LIDAR


Wall
Laser
Beam
D= ct/2
Where:
D – Distance
c- speed of light
t – time of flight
Rotation (around 10 rps) - optional depending on use case and sensor type

LIDAR - Types

- Common types in robotics Types: solid state, rotating
  - Solid state: single chip nothing is rotating
  - Rotating: single laser rotates around

Solid state	Rotating
Relatively new	Well tested
Works over relatively short range	Expensive
Promises to be much cheaper	Long range
	Can be easily damaged

LIDAR - usages

- Robotics:
  - Simultaneous localization and mapping
  - Classification of objects
- Surveying
  - Create accurate maps either 3D or 2D
- Agriculture
  - Show exact plants height and empty spaces
- Inspections
  - To detect damage quickly

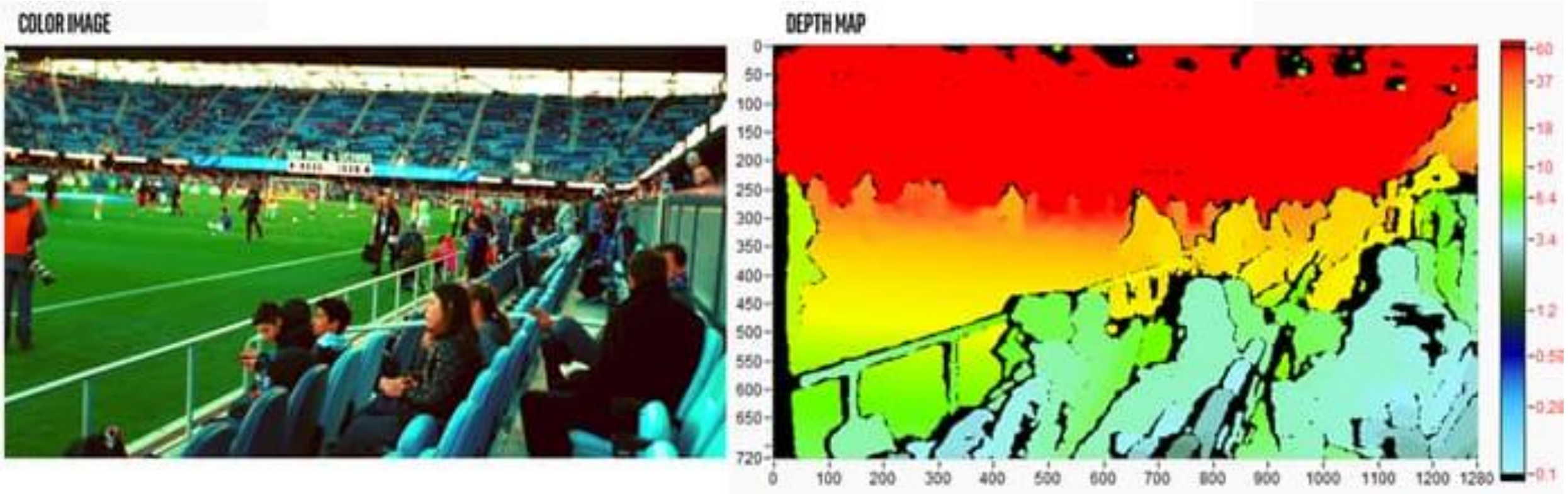
Sensors overview

Cameras – important parameters

- **Field of View (fov):** How wide can camera see
  - Fisheye lense offer extremely large fov, but image is wrapped around the edges
- **Resolution**: how many pixels per image, more pixels mean more accurate image, but also higher computational cost
- **ISO**: how sensitive the sensor is, larger values tend to increase the noise (graininess) of the image

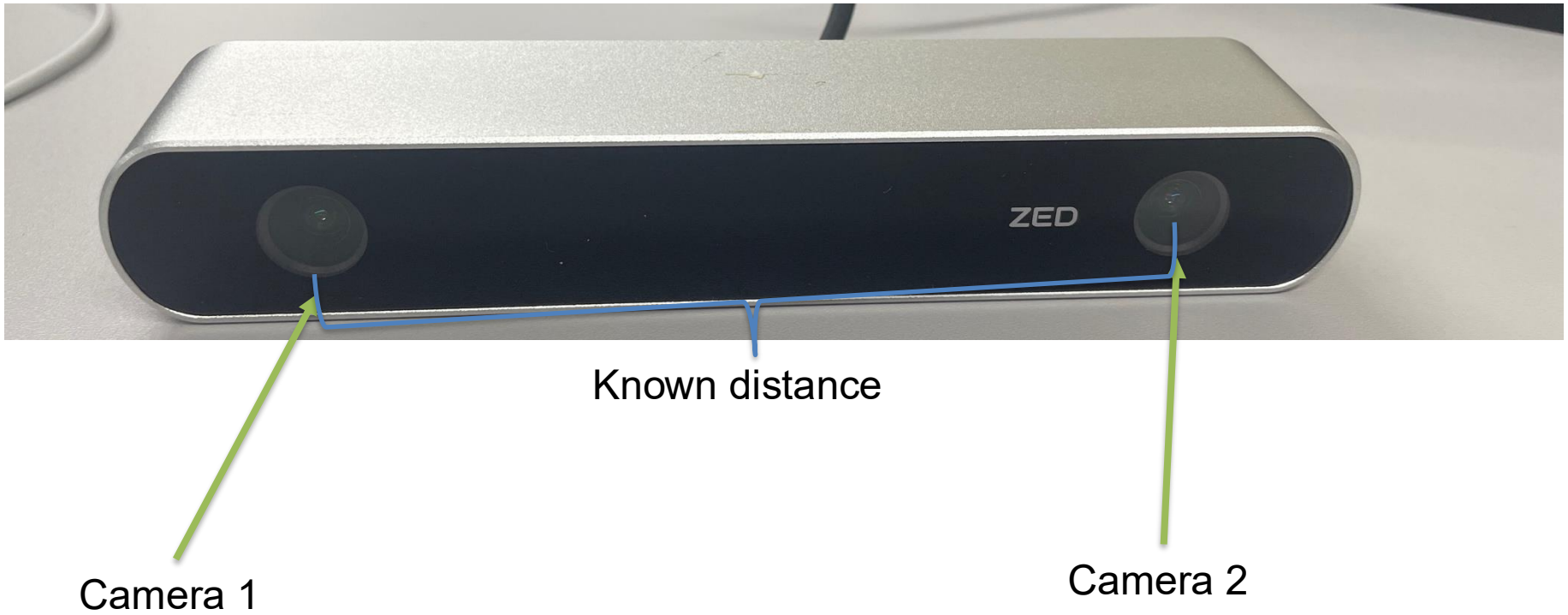
Depth cameras

- Depth camera produces image where pixels are depth values
- Many types, here we briefly cover stereo type



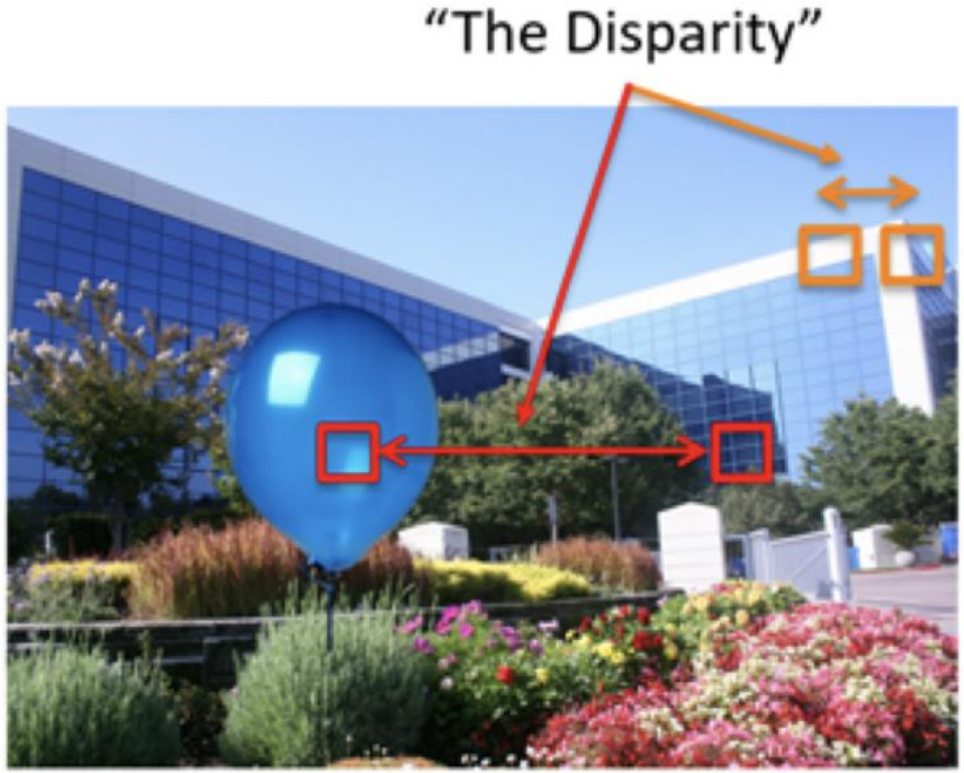
Stereo depth cameras working principle

- Similar in principle how the human eyes work
- Two cameras are placed known distance apart



Stereo depth cameras working principle

Left camera



Right camera

- By knowing disparity and distance between lenses The distance to pixels can be computed (See async material for details)

Stereo depth cameras pros and cons

Pros	Cons
Accurate enough	Does not work well with glass and mirrors(struggles to find good features as image moves)
Can work in any light conditions	Selecting right features to compare is an area of ongoing research
Can work both outdoor and indoor	

Depth Cameras vs LIDAR

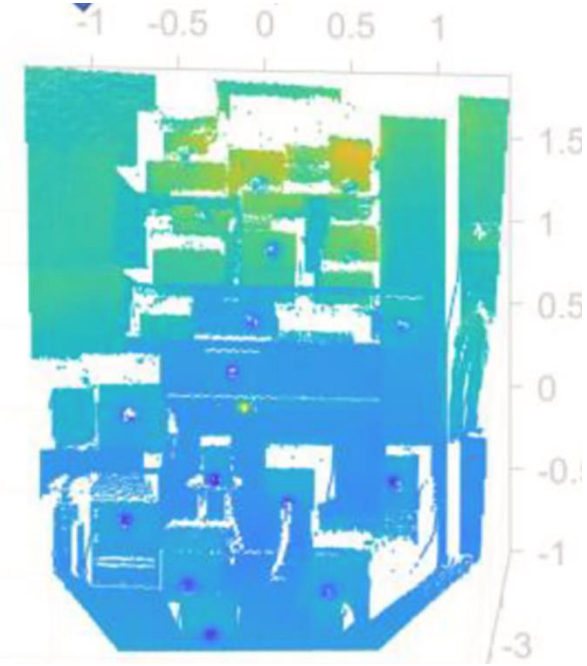
- Both can be used for localization

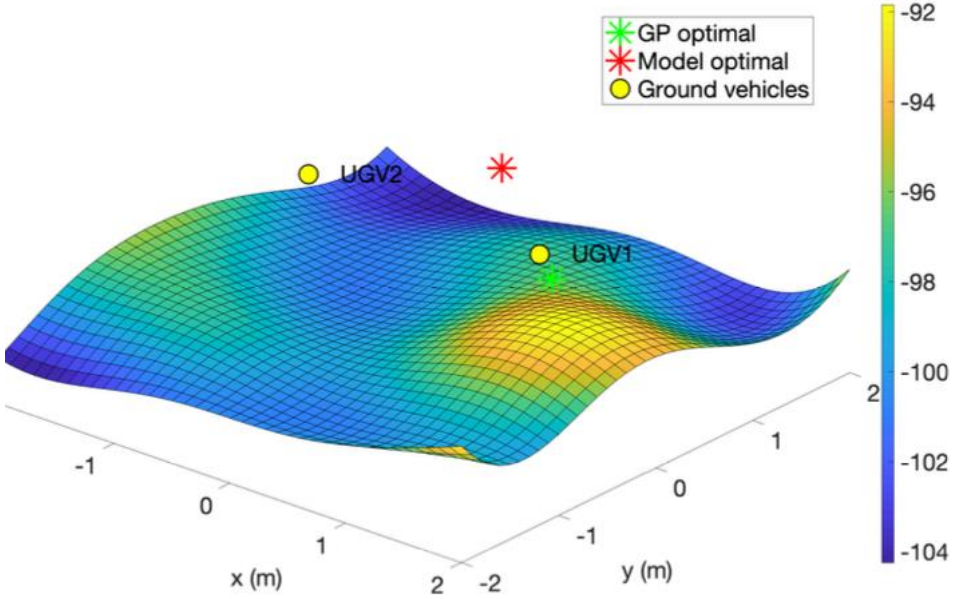
Depth Cameras	LIDAR
Lightweight	expensive
cheap	Very long range
Large field of view (fov)	Large fov LIDARs exists but they are even more expensive
Requires significant computational power for feature selection	More moving parts so reliability can be an issue

Sensors overview

Mission sensors

- They work to fulfil specific mission requirements
- Some examples:





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Summary

- Actuators were introduced
- Motors were discussed in detail
- Locomotion methods in robotics were introduced
- Common sensors were presented and discussed

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Thank you for listening and now onto activity